

Islam Asous

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EDUCATION

Hunter College, City University of New York (CUNY)

New York, NY

Bachelor of Arts in Computer Science, Minor in Mathematics

Expected Dec 2026

- **Relevant Coursework:** Data Structures & Algorithms, Programming Methods, Computer Theory, Computer Architecture, Database Management, Foundations & Practice of Modern AI Systems, Automation Engineering, Linear Algebra

EXPERIENCE

NYC Department of Transportation

Aug 2026 – Present

Software Engineer Intern

Remote

- Scoping and delivering a minimum viable wind load capacity map against a formal project scope and change-order process, translating stakeholder requirements into defined GIS deliverables and timelines.
- Digitizing NYC Building Code §1609.4.3 shoreline wind exposure figures across all five boroughs and classifying building footprints into Exposure B, C, and D using code distance thresholds (600–5,000 ft).
- Communicating progress through written status reports and oral presentations to DOT leadership, practicing stakeholder management and technical communication.

Project Sisypus, Clemson University

June 2026 – Present

Software Engineer REU

Remote

- Maintain a authentication-tracking browser extension (*WXT, Vue 3, TypeScript, Dexie.js*) for NSF-funded research (\$650K, Award #2521037) on authentication-fatigue-driven password reuse across enterprise platforms.
- Built a real-time broadcast notification system (*Vue 3, Pinia, Supabase Realtime*) enabling admins to push instant announcements, enforcing role-based access via Postgres Row-Level Security.
- Embedded *Claude Code* into an agentic development workflow spanning implementation, testing, and cross-repository debugging (Supabase, OAuth, CI), accelerating feature delivery while maintaining 90%+ test coverage.

CUNY Tech Prep

May 2026 – Present

Data Science Fellow

New York, NY

- Selected for a competitive fellowship with students from across the 11 CUNY senior colleges where Fellows create end-to-end data science and AI projects using tools such as Python, SQL, Pandas, scikit-learn, and Hugging Face.
- Participate in weekly courses and learn industry best practices for exploratory data analysis (EDA), feature engineering, statistical analysis, data visualization, LLMs, RAGs, vector databases, and generative AI applications.

Hunter College Computer Science Department

Aug 2025 – Present

Undergraduate Teaching Assistant

New York, NY

- Supported 600+ students across Computer Architecture I & II, Computer Theory, and OOP/Data Structures & Algorithms by leading weekly review sessions and office hours.
- Co-designed quizzes, midterms, and finals for courses of 150–300+ students; reduced grading turnaround by building automated evaluation tools with faculty.

PROJECTS

Gridlock, Multi-Agent Specification-Gaming Testbed

Oct 2026 – Present

AI/ML Engineer — Foundations & Practice of Modern AI Systems(Course)

Remote

- Building a reproducible multi-agent gridworld that measures **specification gaming**—agents satisfying stated rules (R) while violating hidden designer intent (R*)—extending the AI Safety Gridworlds framework with adversarial multi-agent scenarios.
- Train a **tabular Q-learning agent** that discovers rule-legal “loopholes” through trial and error, benchmarked against **A*** path-planning and **constraint-satisfaction** agents behind a shared agent interface to isolate how agent type affects gaming.
- Engineer an **oversight monitor** that scores behavior against hidden intent (R*) and flags divergence, evaluated by detection rate, time-to-detect, and false-positive rate against a non-ML static-rule baseline and a trace-aware vs. action-only ablation.

Token Watching, CUNY Hunter

Aug 2026 – Present

AI/Backend Engineer — Capstone

Remote

- Co-develop **Token Watching**, a Chrome extension that cuts LLM prompt costs in real time inside tools people already use (ChatGPT, Claude), via a three-agent pipeline (Auditor → Optimizer → Validator).
- Accomplish token-waste detection at near-zero marginal cost and measurable prompt savings, as measured by resolving most prompts through deterministic checks alone and cutting a representative prompt 74% (31 to 8 tokens) with intent preserved, by building the **Auditor** (*Python, Tiktoken, Anthropic*) and **Optimizer** agents.
- Accomplish trustworthy, inspectable rejection of meaning-changing rewrites, as measured by a 1–10 LLM-judged quality score with a pass/fail threshold, by building the **Validator** agent around a Read-Answer-Cite-Think reasoning structure that grounds each verdict in quoted evidence from the original prompt.

TECHNICAL SKILLS

Programming: C++, Python3, HTML/CSS, TypeScript, JavaScript, Rust, Swift/SwiftUI, Dart, SQL (PostgreSQL)

Frameworks, Cloud & Tools: Git/GitHub, React, Node.js, Express.js, Flutter, Firebase, Supabase, Docker, REST APIs

Certifications: AWS Skill Builder - Developer, Cloud Operator (SysOps), DevOps (In Progress); CodePath AI201: Applications of AI Engineering (In Progress);